# Vishwash Tank

**Specialization:** Application Security, VAPT, API & Mobile Security, Python Automation, Application Support

## PERSONAL INFORMATION


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## TECHNICAL SKILLS

| | |
|---|---|
| **App Security:** | Web & API Security Testing, Android Security, Secure SDLC, Threat Modeling, Access Control, OWASP Top 10, OWASP API Top 10, MASVS, CVSS v3.1 |
| **Security Tools:** | Burp Suite, OWASP ZAP, Postman, HTTP Toolkit, MobSF, JADX, APKTool, Wireshark, Nmap, Splunk, Elastic Stack, Git |
| **OS & Cloud:** | Linux (Server Administration & Shell Scripting), Docker, Google Cloud Platform (GCP) |
| **Programming:** | C++ (OOPs), Python (Scapy, PyShark, Sockets, Hashlib, Threading), Bash Shell Scripting, SQL |
| **IoT & Hardware:** | ESP8266, ESP32, Arduino, NRF24L01, BLE Security, NTAG213 RFID, RC522, Rubber Ducky HID Injection |

## EXPERIENCE

### Cyber Crime Intern — May 2026 – Present

**Rajasthan Police Headquarters – SP Cyber Crime, Lal Kothi, Jaipur**

- Conducted authorized VAPT on internal web apps and REST APIs — evaluating authentication bypass, IDOR, and broken access controls (OWASP API Security Top 10).
- Reverse-engineered Flutter Android application with MobSF, JADX, and APKTool; identified hardcoded AES-128-GCM keys and cleartext storage via entropy analysis.
- Intercepted mobile API traffic using HTTP Toolkit and Burp Suite; mapped endpoints systematically and validated authorization failures across user roles.
- Documented a critical IDOR vulnerability (CVSS v3.1: 10.0) in an enterprise web application, prepared remediation documentation, and coordinated disclosure via CERT-In.
- Built Python automation scripts for parameter fuzzing, endpoint discovery, and log correlation — accelerating API attack surface enumeration and triage.

### Technical Project Associate — Dec 2025 – Apr 2026

**Vikram Dental Care & National Forensic Sciences University (NFSU), Gandhinagar**

- Engineered Smart Denture NFC Scanner — forensic victim identification system embedding passive RFID/NFC micro-tags into dental prosthetics for disaster identification.
- Programmed Arduino with RC522 modules to test NTAG213 tags; characterized read/write thresholds, data retention under thermal stress, and forensic recovery durability.
- Collaborated with NFSU forensic faculty to design tag memory mapping and ensure cryptographic compliance with forensic evidence admissibility standards.

### Junior Cybersecurity Trainer — Aug 2024 – Feb 2025

**Cyber Octet Pvt. Ltd., Ahmedabad**

- Delivered instructor-led training on practical web security, network traffic analysis with Wireshark, and automated reconnaissance using Nmap.
- Built reproducible CTF lab environments demonstrating OWASP Top 10 vulnerabilities; authored comprehensive technical walk-throughs and student grading rubrics.

## VULNERABILITY DISCOVERIES

**[1]** **Critical IDOR & Broken Access Control — CVSS 10.0.** Identified critical authorization bypass enabling unauthorized data access; responsibly disclosed via CERT-In.

**[2]** **Hardcoded Cryptographic Keys in Mobile App** — Discovered static AES-128 keys in decompiled bytecode via entropy analysis, allowing offline data decryption.

**[3]** **Broken Object Level Authorization (BOLA)** — Identified missing server-side authorization checks across REST API endpoints permitting cross-account data access.

**[4]** **Insecure Cleartext Local Data Storage** — Identified sensitive authentication tokens and user data stored unencrypted in local application storage.

**[5]** **Vertical Privilege Escalation** — Bypassed role validation checks through parameter manipulation, allowing standard accounts to execute administrative actions.

## SELECTED PROJECTS

**NoCTF — Labs-as-a-Service**
Self-built training platform provisioning isolated ephemeral lab environments on demand. GCP backend, Docker-containerized labs, Supabase auth. Patent filed. [**website**]

**WiFi Attack Toolkit (ESP8266)**
Custom C++ firmware for ESP8266 microcontrollers investigating 802.11 management frame vulnerabilities — deauthentication injection, beacon flooding, and probe logging in controlled RF labs. [**GitHub**]

**BLE Attack Device (ESP32 + nRF24L01)**
Dual-radio embedded tool auditing Bluetooth Low Energy and 2.4 GHz protocols. Implements BLE advertisement spoofing, GATT service enumeration, and proximity spoofing on IoT devices. [**GitHub**]

**SOARX Security Dashboard**
Real-time security telemetry and server health monitoring dashboard built with Python. Aggregates host metrics, active daemon states, and threshold alerts for lightweight production lab setups. [**website**]

**Rubber Ducky USB Automation**
Keystroke injection payload suite utilizing Rubber Ducky scripting to assess physical workstation defenses — automated triage droppers, reverse shells, and persistence verification. [**GitHub**]

**SIEM Deployment Lab — Elastic Stack**
Centralized logging infrastructure using Elasticsearch, Logstash, and Kibana. Configured Linux auditd ingestion, Windows Event Forwarding, and Suricata IDS alert pipelines for threat correlation.

## CORE COMPETENCIES

- **Log & Alert Analysis:** Read logs and alerts in Splunk and Elasticsearch to figure out what broke and why.
- **API Debugging:** Postman, Burp, HTTP Toolkit to trace failed requests, odd status codes, bad payloads.
- **Android Pentesting:** MobSF, JADX, APKTool to decompile apps, find hardcoded keys & cleartext storage.
- **Web & API Pentesting:** Burp Suite, ZAP, Nmap for IDOR, BOLA, privilege escalation, OWASP Top 10.
- **Linux & Scripting:** Linux, SSH, Bash & Python scripts for fuzzing, endpoint discovery, log correlation.
- **Monitoring (lab):** Built SOARX in Python to track host health, daemon status and threshold alerts.
- **Docker & Cloud:** Run lab environments in Docker containers on GCP (NoCTF), Git for version control.
- **SQL:** Basic queries and data checks.
- **Reports & RCA:** Findings, root cause and fix steps written so devs can follow (CERT-In, Medium).
- **FDE, Cybersecurity (basic):** Forward Deployed Engineer basics: client-side VAPT, SIEM setup.

## EDUCATION

**Bachelor of Computer Science (B.Sc.)** **2023 – 2026**
**Gujarat University, Ahmedabad • CGPA: 7.48 • First Class with Distinction**

## CERTIFICATIONS

**Advance Diploma in Cyber Security** • **Pursuing: eJPT** — eLearnSecurity Junior Penetration Tester

## TECHNICAL WRITING

Publishes technical cybersecurity research, vulnerability disclosure methodologies, and Android reverse engineering write-ups on **Medium (@vishwashtank)** — written for practitioners and engineering teams. [**Medium**]

## LANGUAGES

**English** (Professional) • **Hindi** (Native) • **Gujarati** (Native) • **Italian** (Learning)

## PROFILE

Security and support practitioner with grounded expertise in application security, API troubleshooting, and Linux system automation. Dedicated to reproducible diagnostics, root-cause isolation, and secure engineering practices. Proven self-starter with strong ownership mindset and rapid adaptability to production stacks.